

Deep Text Mining of COA Annual Audit Reports for Systemic Financial Misappropriation

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Abstract

The Philippine Commission on Audit (COA) releases detailed annual audit reports, but finding evidence of financial misuse in these long, messy PDFs is difficult. This proposal asks if a document-level Transformer model can automatically find connections between a violation, the monetary amount involved, and the responsible person or organization. We plan to build a dataset from COA reports, manually check a small portion of it, and fine-tune a RoBERTa model to perform named-entity recognition (NER), violation classification, and relation extraction. We will also use layout and section features to help the model read better than a simple keyword search. We will evaluate our model against basic keyword and sentence-level baselines. The main goal of this project is to create a tool that helps human auditors review documents faster, not to automatically declare legal guilt.

Keywords

Natural Language Processing, Document Understanding, Named Entity Recognition, Financial Auditing, Philippine Government

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1 Introduction

COA annual audit reports are important public records, but they are not used enough. A single report can be hundreds of pages long and contain issues about procurement, cash advances, and contractors. Right now, human reviewers have to manually connect a violation in one paragraph to a peso amount in a table and a contractor named somewhere else. Because the information is scattered, it is hard to track recurring problems across different local government units (LGUs).

The research question for this project is: Can a document-level Transformer extract meaningful financial violation details from COA reports more accurately than basic keyword searches or sentence-level models?

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Our approach uses text extraction, page coordinates, section segmentation, and multi-task learning. We need this because the meaning of words depends on the context (for example, "not liquidated" means different things depending on who it refers to), and we need to link entities together, not just find keywords.

Our expected contributions are:

- a clean, ready-to-use dataset of COA reports;
- a multi-task machine learning model that extracts violation-entity-amount links;
- an evaluation process to see if the model works across different LGUs and years; and
- a simple interface to rank findings for human auditors.

2 Related Work

Pre-trained language models are the foundation of this project. BERT introduced bidirectional Transformers and showed great results for token and sentence classification tasks [4]. Early work by Lample et al. showed that neural networks could do NER well without relying on strict rule-based lists [8]. For processing document images, LayoutLM uses both the text and the 2D position of words on a page, which is very helpful for reading forms and reports [15]; LayoutLMv2 improves this by adding spatial attention [16]. However, COA reports are very long, so we also need models that can understand context across many sentences.

The DocRED paper points out this exact problem: finding relationships often requires reading multiple sentences and entities in a document [17]. Long-document models like Longformer [3] provide a good way to handle this, but they aren't trained for financial audits.

Working with government financial text also has specific vocabulary challenges. Liu et al. showed with FinBERT that standard models struggle with financial terms, meaning we need domain-specific training [10]. Other recent papers highlight how NLP can be used to find corruption in government procurement [1]. While standard NER is good at finding names and amounts [11], there is still a gap in applying these models to long Philippine COA audit reports to automatically find complex financial violations.

3 Proposed Methodology

Figure 1 shows our project pipeline. First, we will parse each PDF to get the text, page numbers, bounding boxes, tables, and headers. If a page is scanned and has no text, we will use OCR to read it. We will clean the extracted text by fixing spacing and currency symbols, but we will keep track of where the text came from on the original page.

Since standard Transformers can only read 512 tokens at a time, we will split the reports into chunks of 384 tokens, with a 64-token

overlap so we don't cut sentences in half. We will use RoBERTa-base as our main encoder because it performs better than the original BERT [9]. The chunk representations will be combined using a two-layer document Transformer to help the model understand long texts [12]. The NER part of the model will look for specific tags: PERSON, ORGANIZATION, LGU, CONTRACTOR, AMOUNT, DATE, PROJECT, and VIOLATION.

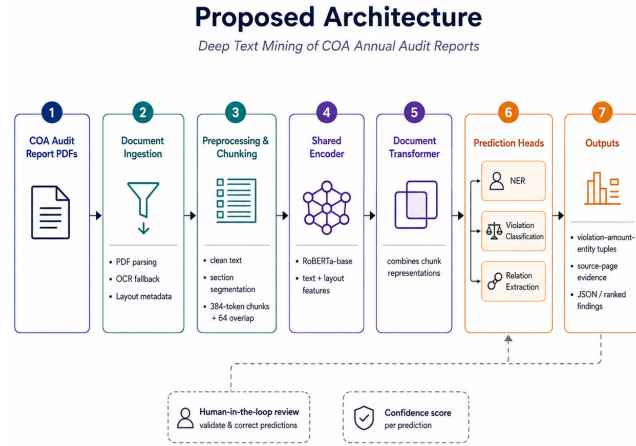


Figure 1: Proposed architecture. Shared document representations feed three supervised heads; all predicted tuples retain source-page and sentence evidence.

The model will also classify violations into five categories: unauthorized expenditure, unliquidated cash advance, procurement irregularity, unsupported disbursement, and suspicious contractor activity. To find relationships between entities, we will use a scoring model [5] to predict links like INVOLVES, AMOUNT_OF, or RESPONSIBLE_FOR. Doing all these tasks together in one model helps prevent errors that usually happen when tasks are done one by one [2].

For our data, we will use a two-step annotation process. First, we will use simple code rules to find obvious dates and currency amounts. Then, two human annotators will manually check and label the entities and relationships in 250 reports. The rest of the reports will be labeled automatically using strong rules to help pre-train the model [13]. The final test set will be completely hand-labeled.

Output schema and decoding. The final output for each report will be a JSON file with the document details, entities, violations, and relationships. During testing, the model will only look for relationships between entities that are within 12 sentences of each other.

Leakage and uncertainty controls. LGUs often copy-paste the same text from year to year. To prevent data leakage (where the

model memorizes the text instead of learning), we will group the train/test splits by LGU and year. We will also make sure the model outputs a confidence score for its predictions, so human reviewers know which findings might be uncertain.

Analysis protocol. We will report our results with 95% confidence intervals and do an error analysis to see where the model gets confused (for example, if OCR errors caused the mistake). We will also do a small user study with three reviewers to see if the model actually helps them review documents faster.

4 Experimental Plan

Dataset. We plan to download around 1,000 searchable COA annual audit report PDFs from 2018 to 2023, covering at least 100 LGUs. We will split the data into 70% training, 15% validation, and 15% testing. We will split them by LGU-year to make sure the model doesn't just memorize specific local governments. Preprocessing will include PDF text extraction, OCR if needed, and sentence splitting.

Baselines. We will compare our model against four baselines: (B1) a traditional TF-IDF and SVM model [7] to see if deep learning is even necessary; (B2) a basic RoBERTa model that only looks at single sentences; (B3) a Longformer model that handles long text but ignores the page layout; and (B4) a LayoutLMv3 model to test if knowing the visual layout helps.

Hyperparameter tuning. We will tune the learning rate, batch size, chunk length, and dropout. We will run the model with five different random seeds and choose the best version based on the validation F1 score.

Ablations. To see which parts of our model are actually useful, we will try removing components one by one, like the layout features, document-level attention, or the OCR fallback [6].

Evaluation metrics. Following standard NLP practices [14], we will evaluate the NER using precision, recall, and F1 scores with exact span matching. For the overall project, our main metric will be the end-to-end tuple F1 score (did it correctly link the violation, amount, and entity together?). We will also measure how fast the model runs.

Implementation. We will write the code using PyTorch and Hugging Face Transformers. We will use pdfplumber and Tesseract for the PDF extraction. We will run the experiments on an NVIDIA RTX 4090 GPU. Training should take about 4 to 8 hours.

5 Risks, Ethics, and Reproducibility

Even though COA reports are public documents, we still need to be careful with names and financial allegations. We will present this project as a search tool to help auditors, not as an AI that decides if someone broke the law. We will make sure the model points to the exact sentence it used for its prediction. Expected risks include OCR mistakes and changing report formats, which we will try to fix during preprocessing. We will release our code and dataset splits if the data distribution rules allow it.

6 Expected Outcome

This project will test if a Transformer model can turn long, messy COA reports into a clear, connected graph of financial findings. We will consider the project a success if our model gets a significantly higher F1 score than the baseline models and helps human auditors

find important information faster. Even if the model doesn't beat the baselines, our error analysis will still show what makes COA reports so hard for NLP models to read, which can help future research.

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